Kuang Ye

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AI Application Engineer
Embedded Systems Engineer
Optoelectronic Systems Engineer

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Professional Summary

Final-year optoelectronic engineering undergraduate with hands-on experience in automotive electronics testing and a delivery record spanning **embedded systems**, **AI tooling**, and **optical design**. Develops from schematic and PCB design through **STM32** firmware to edge deployment of **YOLO-family** vision models. Applies AI coding agents to cut engineering turnaround **substantially**, with national- and provincial-level awards in optoelectronic design and mathematical modeling.

Technical Skills

- **Embedded & Hardware:** STM32 (ARM Cortex-M) firmware in C/C++, circuit design, PCB layout, sensor interfacing, motor driver tuning, SMT hand soldering, oscilloscope debugging
- **AI & Programming:** Python, C, C++; AI coding agents (Claude Code, Codex, Kimi Code); prompt engineering, retrieval-augmented generation (RAG), tool/function calling and reusable skill design; YOLO-family model development and edge deployment
- **Engineering Software:** Keil µVision 5, JLCPCB EDA, SolidWorks, AutoCAD, Proteus, NI Multisim
- **Optical & Simulation:** Zemax optical design, LabVIEW; applied optics, photoelectric detection
- **Media & Documentation:** Video editing and photography (contributing photographer, VCG); self-produced product demos, marketing collateral, and technical illustrations
- **Languages:** Mandarin (native), English (fluent), Cantonese (fluent)

Education

Changchun University of Science and Technology (CUST) — Bachelor of Engineering, Information Countermeasure Technology — **Sep 2023 – Jun 2027 (expected)**

School of Optoelectronic Engineering; ranked among the **top students** in the program; recipient of a university scholarship and multiple provincial-level competition awards

- **Core coursework:** Electronic Technology, Applied Optics, Signals and Systems, Communication Principles, Embedded System Design, Optical Design, Electronic Warfare Technology, Electro-Optical Countermeasure Technology, Photoelectric Detection Technology

English Proficiency

- **CET-4** (College English Test, Band 4): **480 / 710** — Passed • **CET-6** (College English Test, Band 6): **400+ / 710**

Experience

Changchun Jieyi Automotive Technology Co., Ltd. — Engineering Assistant — **Sep 2026 – Present**

New-Energy Vehicle Electronics and Wiring Harness Testing, Changchun, China

- Execute testing and development tasks on new-energy-vehicle electronic hardware and **full-vehicle wiring harnesses** across **environmental, durability, and high/low-voltage** test categories.
- Own **end-to-end test cycles**: perform test runs, record data, and issue inspection reports, while designing and iterating test procedures to meet project requirements.
- Support project engineers in functional development by aligning test plans, locating faults, tracking test milestones and delivery progress, and feeding results back to development to close the loop.
- Contributed to the development of an experiment-report management system backed by **SQL**, standardizing how test reports are archived, retrieved, and traced across projects.
- Participated in building an **AI-assisted knowledge base** and workflow automation for the laboratory, consolidating test know-how and standardizing recurring procedures.
- Introduce **AI tools** into daily test and documentation workflows for data processing and process mapping, shortening routine turnaround and building a library of reusable working methods.
- Produce test records, reports, and data summaries with Word, Excel, and PowerPoint.

Projects

"Sun-Chaser" Dual-Axis Tracking System for Offshore Photovoltaic Plants

Project Lead — 14th National University Optoelectronic Design Competition (Creative Division) **Mar 2026 – Jun 2026**

Technologies: LiTaO3 pyroelectric sensing, dual-axis closed-loop control, SolidWorks structural design, IP68 enclosure

- Led the full project lifecycle for a dual-axis tracking system proposed for offshore PV, targeting the efficiency losses of fixed installations and the weak environmental adaptability of existing trackers.
- Ran market research, technology benchmarking, and competitive analysis to define a differentiated position, then proposed a **dual-axis closed-loop** tracking architecture based on the **LiTaO3 pyroelectric** effect.
- Authored the feasibility case demonstrating a generational advantage over imported pyroelectric devices and domestic photoelectric solutions across four dimensions: broad spectral response (**300–2500 nm**, covering **~96%** of solar irradiance), **millisecond-level** response, fully passive structure, and **IP68** marine-grade sealing.
- Defined the core technical metrics matrix and wrote the system design specification.

STM32 Multifunction Electronic Piano

Independent Course Project — Outstanding Course Project Award; code, schematics, and debug logs published on GitHub **Sep 2025 – Dec 2025**

Technologies: STM32 MCU, PWM audio synthesis, RC low-pass filtering, SMD soldering, oscilloscope measurement

- Designed and built a multifunction electronic piano on **STM32**, driving a passive buzzer through PWM output and converting the PWM square wave to analog audio with a self-designed **RC low-pass filter**.
- Calibrated PWM frequency and duty cycle against the **twelve-tone equal temperament** scale and validated the result through physical performance testing.
- Hand-soldered **SMD** components, measured PWM waveform distortion on an oscilloscope, and iterated filter capacitor values to improve audio quality, maintaining **hardware debug logs** that tracked every parameter iteration.

RoboMaster Combat Robot Platform

Mechanical & Electronic Integration Team Member **Mar 2024 – Jun 2024**

Technologies: SolidWorks 3D modeling, lightweight structural iteration, mechanical-electrical interface design

- Co-designed infantry and hero robots, owning 3D modeling of chassis and gimbal structures, key component selection, and multiple **lightweighting iterations** in **SolidWorks**; delivered several assembly proposals that passed internal review.
- Aligned mechanical-electrical interfaces and cable routing with the embedded team, supported prototype assembly and on-site tuning, and consolidated the results into a reusable chassis and gimbal design for future seasons.

AI Agent Engineering Workflows

Personal Project Portfolio **Jun 2024 – Present**

Technologies: Claude Code, Codex, Kimi Code, prompt engineering, RAG (vector retrieval), tool/function calling, Python, Shell

- Designed reusable **prompt-engineering playbooks** covering role definition, few-shot exemplars, structured output constraints and iterative refinement, stabilizing agent output across code generation, bug localization and technical writing.
- Built a local **retrieval-augmented generation (RAG)** workflow indexing course notes, datasheets and project documents, grounding agent output in retrieved context and cutting hallucinated APIs and repeated source-hunting.
- Wired **tool calling and function calling** into the workflows so agents execute shell scripts, run Python and call external APIs instead of only returning text, then packaged recurring procedures into **reusable skills** for one-command execution.
- Applied the stack to embedded and optical work: generating STM32 peripheral initialization code and drafting Zemax and LabVIEW simulation and data-processing scripts; reduced average preparation time for coursework and competitions by **approximately 50%**.

Honors & Awards

- **Second Prize**, University Round — 14th National University Optoelectronic Design Competition, Creative Division (2026)
- **First Prize**, Jilin Regional — China Undergraduate Mathematical Contest in Modeling (2025)
- **First Prize** — University Mathematical Modeling Competition, awarded multiple times (2024)
- **Second Prize** — University Innovation and Entrepreneurship Competition (2024)
- **Second-Class University Scholarship** (2023)
- **Outstanding Course Project** — awarded repeatedly for Applied Optics, Embedded Systems, and Integrated Optoelectronics projects; core embedded projects published on GitHub